

From Accurate Forecasts to Feasible Plans: Feasibility-Constrained Model Selection for Operational Demand Planning

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Abstract

Operational demand planning increasingly relies on predictive models, but operations execute planning signals rather than forecast accuracy metrics. A numerically accurate forecast can still produce volatile inventory targets, frequent switching, or execution violations when downstream systems cannot absorb abrupt plan changes. This paper formulates forecast-driven demand planning as feasibility-constrained model selection. Candidate forecasts are converted into executable planning signals and evaluated using WAPE, inventory exposure, planning volatility, switching burden, execution violations, normalized planning loss, greedy-to-DP gap, and oracle gaps. DataCo execution records provide data-informed execution-risk scenarios, Favorita retail demand data provide the main forecast-to-plan proof of concept, and M5 retail data test hierarchy and intermittent-demand robustness. Across 123 paper-facing evaluation groups, the accuracy-first deployable strategy differs from the operational-loss-optimal deployable strategy in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. Favorita results show ranking shifts under severe execution-risk scenarios, while M5 results show that planning grain and intermittency change the preferred deployable mechanism. The contribution is not a new forecasting model. It is an interpretable operations decision layer showing that forecast accuracy should be treated as a diagnostic metric, while deployment should be evaluated by feasibility-adjusted planning utility.

Keywords

Demand planning, Forecast-to-plan conversion, Operational feasibility, Planning stability, Execution risk.

1 Introduction

Industrial demand-planning systems increasingly rely on predictive models to support replenishment, inventory positioning, labor planning, transportation commitments, and capacity allocation. Recent reviews show that artificial intelligence and machine learning are becoming more visible in supply-chain planning and demand-planning practice (Walter et al. 2025; Culot et al. 2024; Vlachos and Reddy 2025). Forecasting benchmarks and time-series foundation models further show that predictive capability is advancing rapidly (Makridakis et al. 2022; Das et al. 2024; Ansari et al. 2024). These advances create value, but they also sharpen an operations-management problem: better forecast accuracy does not automatically mean better deployment quality.

Operations execute planning signals, not forecast accuracy metrics. A planner cannot execute WAPE or RMSE. The operation executes an inventory target, a replenishment order, a staffing plan, a production target, or a transportation commitment. When a forecast-driven plan changes sharply, downstream systems may need supplier coordination, warehouse rebalancing, labor rescheduling, or enterprise planning-system updates. These changes can be costly even when the forecast itself is numerically accurate.

This creates a planning-execution gap. Forecasting models may adapt daily or weekly, while procurement lead times, replenishment workflows, planner governance, and execution infrastructure may absorb only gradual change. A forecast-driven plan can therefore create planning volatility, model-switching burden, and execution-capacity violations. Standard forecasting evaluation often emphasizes WAPE, MAE, RMSE, or related accuracy metrics, while deployment requires feasibility-adjusted planning utility.

This paper studies feasibility-constrained model selection for operational demand planning. The central thesis is that forecasting-driven strategies should be selected by operational feasibility as well as prediction accuracy. Candidate

forecasts are converted into planning signals, evaluated against inventory exposure and execution constraints, and compared through deployable decision-layer policies. The empirical contribution is to measure how often and by how much accuracy-first selection differs from operational-loss-optimal deployable selection.

The study uses three public-data roles. DataCo supply-chain execution records are used to anchor execution-risk scenarios from observed late-delivery heterogeneity. Favorita daily retail demand data provide the main forecast-to-plan proof of concept. M5 retail data provide hierarchy and intermittent-demand robustness. These datasets are used to evaluate whether the planning-execution gap remains visible under different operational conditions; they are not used to claim that one forecasting model universally dominates.

The paper contributes to IEOM-oriented operations research in four ways. First, it defines a forecast-to-plan evaluation layer that reports forecast accuracy separately from operational planning utility. Second, it operationalizes the planning-execution gap through execution violation rate and planning-execution gap rate. Third, it compares accuracy-first, ensemble, smoothing, greedy, dynamic programming, and budgeted dynamic programming deployable strategies. Fourth, it validates the tradeoff using DataCo-informed scenarios, Favorita proof-of-concept results, and M5 hierarchy/intermittency robustness.

1.1 Objectives

The specific objectives of this study are:

1. Define a forecast-to-plan evaluation layer that measures operational feasibility in addition to WAPE.
2. Operationalize the planning-execution gap using execution violation rate and planning-execution gap rate.
3. Use DataCo execution records to construct data-informed execution-risk scenarios.
4. Compare accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP deployable planning strategies.
5. Validate the planning-execution gap across Favorita and M5 and translate the findings into practical planning recommendations.

2 Literature Review

2.1 AI-enabled Demand Planning and Forecasting Benchmarks

Recent supply-chain research shows increasing interest in AI-enabled demand planning. Walter et al. (2025) review artificial intelligence in demand planning and identify AI tools, supply-chain applications, and digital supply-chain impacts as major knowledge clusters. Culot et al. (2024) review empirical AI studies in supply-chain management and emphasize the need to connect AI adoption to operational research directions. Vlachos and Reddy (2025) similarly review machine learning in supply-chain management and identify planning, execution, and control as areas where machine learning can support decision making.

Forecasting benchmarks and time-series foundation models show the same trend from a forecasting perspective. The M5 competition demonstrates the scale and hierarchy of modern retail forecasting benchmarks (Makridakis et al. 2022). Recent time-series foundation models, including decoder-only forecasting models and pretrained probabilistic time-series models, show that forecasting pipelines are moving toward broader zero-shot and cross-domain capability (Das et al. 2024; Ansari et al. 2024). These studies motivate the forecasting layer of the present paper, but they do not remove the need to evaluate whether forecast-driven planning signals are operationally executable.

2.2 Forecast-to-decision and Predict-then-optimize Research

Forecast-to-decision research is directly relevant because it explains why prediction accuracy and downstream decision quality can differ. Elmachoub and Grigas (2022) show that the standard predict-then-optimize workflow can

be improved by training predictions toward decision loss rather than prediction loss alone. Mandi et al. (2024) survey decision-focused learning and describe how machine learning and constrained optimization can be integrated to improve decision quality under uncertainty. Jia et al. (2025) propose a scenario predict-then-optimize method for data-driven online inventory routing, showing that downstream routing and replenishment decisions require more than point forecast accuracy.

The present paper is related to this stream, but its purpose is different. It does not introduce a new end-to-end learning algorithm. It proposes an interpretable forecast-to-plan decision layer for operational demand planning, where existing candidate forecasts are evaluated by inventory exposure, planning volatility, switching burden, and execution-capacity violations.

2.3 Inventory Implications and Planning Stability

Inventory theory provides the operational foundation for holding and shortage costs. Silver et al. (1998) and Axsater (2015) describe inventory planning as a cost and service-level problem rather than a pure prediction problem. Forecast accuracy metrics remain important, but Hyndman and Koehler (2006) show that forecast-error measures have known limitations and must be interpreted carefully across series and contexts. Forecast combination is also relevant because weighted or averaged forecasts can reduce dependence on a single model (Bates and Granger 1969).

Modern forecasting candidate sets often include nonlinear machine-learning methods such as random forests, XG-Boost, and LightGBM (Breiman 2001; Chen and Guestrin 2016; Ke et al. 2017). These models can improve forecast accuracy, but model deployment also creates governance, monitoring, and switching burden. From an operations perspective, Forrester (1961) shows that industrial systems can amplify instability when material and information flows respond with delays. Advanced planning systems similarly separate forecasting, planning, coordination, and execution layers (Stadtler et al. 2015). This separation motivates explicit measurement of planning volatility and execution-capacity violations.

Rolling-horizon decision theory also supports finite-horizon selection logic because a locally attractive action can be undesirable once future transitions are considered (Sethi and Sorger 1991). This perspective is consistent with the paper's use of greedy-to-DP comparisons to measure myopic planning suboptimality.

2.4 Research Gap

Existing studies improve forecasts, combine forecasts, or optimize downstream decisions. Fewer studies provide a practical decision layer that compares forecasting-driven strategies by inventory exposure, planning volatility, switching burden, and execution-capacity violations. This paper addresses that gap by defining planning-execution gap rate and feasibility-adjusted normalized planning loss. The goal is not to replace forecasting metrics, but to position WAPE as a diagnostic measure alongside operational feasibility.

3 Methods

The proposed method follows the IEOM sample-paper pattern of a framework architecture followed by compact stepwise logic. Figure 1 presents the full decision architecture. Candidate forecasts or planning policies enter the system first, but the decision layer evaluates their implied planning signals before choosing a deployable planning strategy. This framing follows the forecast-to-decision logic that downstream objectives may differ from forecast-error objectives (Elmachtoub and Grigas 2022; Mandi et al. 2024).

3.1 Feasibility-constrained Planning Model

Let i index a planning unit, t index a planning period, and m index a candidate forecasting model. The candidate forecast is $\hat{y}_{i,t}^m$, and the executable planning signal is $p_{i,t}^m$. The planning signal is created through a forecast-to-plan

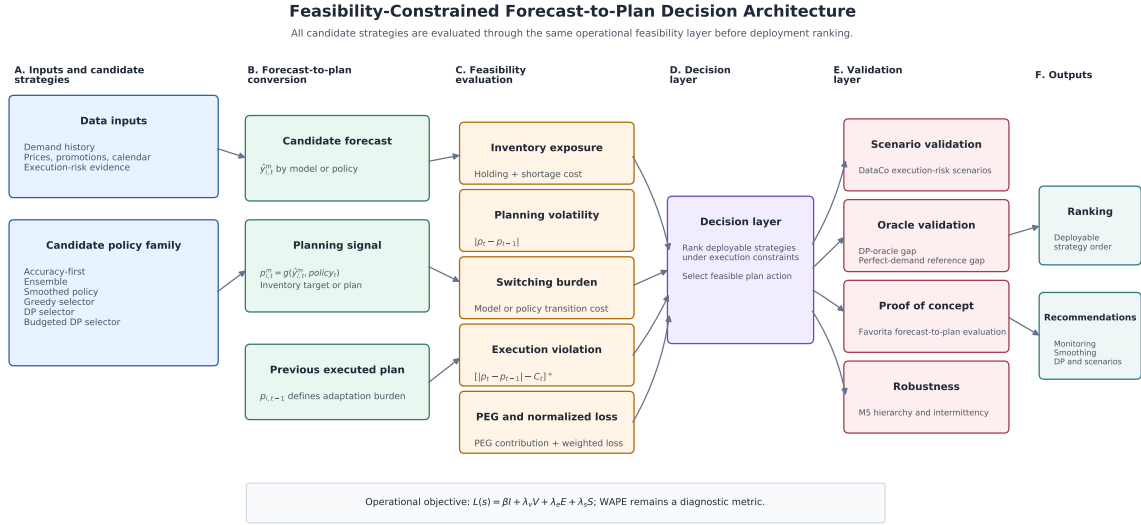


Figure 1: Framework architecture for feasibility-constrained demand-planning selection.

conversion rule:

$$p_{i,t}^m = g(\hat{y}_{i,t}^m, \text{policy}_t). \quad (1)$$

In the current implementation, $p_{i,t}^m$ is an inventory target implied by the forecast and safety-stock policy.

Inventory exposure is measured through holding and shortage terms:

$$IC_{i,t}^m = h[p_{i,t}^m - y_{i,t}]^+ + b[y_{i,t} - p_{i,t}^m]^+, \quad (2)$$

where h is holding cost, b is shortage cost, $y_{i,t}$ is realized demand, and $[x]^+ = \max(x, 0)$.

Planning volatility is the absolute movement in the planning signal:

$$PV_{i,t}^m = |p_{i,t}^m - p_{i,t-1}^m|. \quad (3)$$

Execution capacity $C_{i,t}$ is the amount of plan movement the operation can absorb in period t . The execution violation is:

$$EV_{i,t}^m = [|p_{i,t}^m - p_{i,t-1}^m| - C_{i,t}]^+. \quad (4)$$

The paper-facing planning-execution gap rate for strategy s is:

$$PEG(s) = \frac{1}{T} \sum_{t=1}^T \mathbb{I}\{EV_t(s) > 0\}. \quad (5)$$

The normalized planning loss is:

$$L(s) = \beta_{\text{inventory}} \cdot \text{NIC}(s) + \lambda_{\text{volatility}} \cdot \text{NVC}(s) + \lambda_{\text{execution}} \cdot \text{NEC}(s) + \lambda_{\text{switch}} \cdot \text{NSC}(s), \quad (6)$$

where NIC, NVC, NEC, and NSC denote normalized inventory, volatility, execution, and switching components. WAPE is reported as diagnostic forecast accuracy. The main operational objective excludes direct forecast-error terms because inventory exposure and feasibility terms capture the deployable consequences of forecast-driven planning decisions.

3.2 Feasibility-constrained Selection Procedure

Algorithm 1. Feasibility-constrained selection procedure.

1. Generate candidate forecasts or planning policies for each planning unit and period.
2. Convert each candidate forecast into a planning signal.
3. Compute inventory exposure relative to realized demand.
4. Compute planning volatility and switching burden relative to the previous executed plan.
5. Compute execution violation under the available execution capacity.
6. Aggregate the normalized planning loss.
7. Select the deployable planning action using one of the decision policies.
8. Validate the selected strategy against scenario and oracle benchmarks.

The procedure is modular. Accuracy-first baselines, ensembles, smoothing rules, greedy selectors, DP selectors, and budgeted-DP selectors all pass through the same feasibility layer. This makes their outputs comparable as planning actions, not merely as forecast-error scores.

Forecast-to-plan conversion matters because operations execute inventory targets, staffing signals, replenishment quantities, or other planning actions rather than WAPE values. Planning-execution gap rate is therefore a key metric: it measures how often the selected plan exceeds the adjustment capacity of the operating system. The feasibility layer is the core contribution because it translates model outputs into inventory exposure, volatility, switching burden, and execution violations before ranking strategies. The decision layer is consequently selecting the most deployable planning action under stated constraints, not simply the most accurate forecast.

Table 1 summarizes the decision-layer strategy families used in the empirical evaluation.

Table 1: Decision-layer strategy families and operational interpretation.

Strategy family	Main decision logic	Operational purpose
Accuracy-first	Selects the candidate with strongest validation or test-period forecast accuracy.	Conventional benchmark for WAPE-led deployment.
Ensemble	Combines candidate forecasts using simple or operational-loss-based weights.	Reduces dependence on one model and can dampen abrupt plan movement.
Smoothing	Blends the candidate plan with the previous executed plan.	Models gradual operational adaptation under execution limits.
Greedy	Minimizes current-period operational cost.	Tests whether one-step feasibility selection is sufficient.
DP	Minimizes cumulative finite-horizon operational cost.	Quantifies the value of planning ahead under transition effects.
Budgeted DP	Applies DP while limiting total model switches.	Represents governance constraints on model changes.
Oracle benchmarks	Use non-deployable realized inventory or realized demand information.	Provide diagnostic upper-bound references, not deployment policies.

3.3 Greedy and Dynamic Programming Selectors

The greedy selector minimizes current-period operational cost only. Let $c_t(m_{t-1}, m)$ denote the validation-derived stage cost of switching from previous model m_{t-1} to candidate model m at time t . The greedy selector chooses:

$$m_t^G = \arg \min_{m \in \mathcal{M}} c_t(m_{t-1}, m). \quad (7)$$

This policy is interpretable but myopic because it ignores future switching and execution consequences.

The finite-horizon DP selector accounts for future operational costs:

$$V_t(m_{t-1}) = \min_{m \in \mathcal{M}} \{c_t(m_{t-1}, m) + V_{t+1}(m)\}, \quad (8)$$

with terminal condition:

$$V_{T+1}(\cdot) = 0. \quad (9)$$

The greedy-to-DP gap measures myopic selection suboptimality. DP is included to quantify the value of planning ahead over one-step operational selection.

3.4 Budgeted DP Selector

Some planning organizations impose governance limits on model changes. The budgeted DP selector adds:

$$\sum_{t=2}^T \mathbb{I}\{m_t \neq m_{t-1}\} \leq K. \quad (10)$$

The DP state includes the previous model and the number of switches already used. Transitions that exceed the switch budget are discarded. If missing candidate availability leaves no feasible path, the implementation uses a flagged incumbent-stays fallback: it chooses the first period's lowest-cost model and holds that incumbent when available. This fallback is budget-compliant when the incumbent model is available throughout the horizon and is marked diagnostically if availability fails.

3.5 Ensemble and Smoothing Policies

The Simple Ensemble averages candidate forecasts without using operational-loss information. The Operational-Loss Ensemble uses validation normalized planning loss to assign nonnegative weights. These policies test whether soft forecast composition can reduce abrupt execution burden while retaining useful accuracy.

The smoothing policy changes the executed planning signal, not the underlying forecast:

$$p_t = \alpha p_t^{\text{candidate}} + (1 - \alpha) p_{t-1}. \quad (11)$$

Lower α values reduce execution burden by adapting gradually, but they can increase inventory exposure if the plan responds too slowly to demand changes.

3.6 Oracle Benchmarks and Validation Gaps

The Realized-Inventory Oracle DP is non-deployable. It uses realized test-period inventory outcomes over the same forecast candidates and is used only to quantify the value of unavailable inventory-outcome information. It is not a perfect-forecast oracle.

The Perfect-Demand Oracle uses realized demand as the forecast signal. It is also non-deployable and can still create volatility or execution penalties because perfect demand tracking may move faster than execution capacity. The metric `gap_to_dp_oracle` measures selection suboptimality relative to the realized-inventory DP benchmark. The metric `gap_to_perfect_oracle` measures the total gap to a realized-demand reference, including both forecast uncertainty and feasibility effects.

Table 2 links each method component to its evidence and operational implication.

Table 2: Method components, validation evidence, and operational implication.

Method component	What it tests	Paper-facing evidence	Operational implication
Forecast-to-plan conversion	Whether forecasts become materially different planning signals	Accuracy-first mismatch rate and Favorita frontier	Evaluate plans, not forecasts alone.
Planning-execution gap rate	Whether plans exceed execution capacity	PEG rate: 15.4% accuracy-first vs. 2.7% best deployable	Monitor feasibility as a deployment metric.
DataCo scenarios	Whether empirical execution-risk regimes reorder strategies	Baseline-to-severe Favorita ranking shifts	Use scenario tests before rollout.
Greedy selector	Whether one-step selection is sufficient	Greedy-to-DP gap	Greedy policies can be myopic.
Finite-horizon DP	Value of planning ahead under transition costs	Greedy-to-DP gap and DP oracle gap	Account for future switching and execution burden.
Budgeted DP	Governance limit on model switching	Hard switch budget and flagged fallback semantics	Use when model changes require approval.
Smoothing policy	Gradual operational adaptation	Lower execution burden with inventory tradeoff	Adapt slowly when infrastructure is constrained.
Ensemble policy	Soft composition across models	Lower execution burden in Favorita severe scenario	Reduce model-specific plan jumps.
Oracle benchmarks	Value of unavailable future information	gap_to_dp_oracle and gap_to_perfect_oracle	Separate selection suboptimality from forecast uncertainty.
M5 robustness	Sensitivity to planning grain and intermittency	Best method changes across M5 slices	Avoid one-size-fits-all deployment rules.

4 Data Collection and Experimental Design

The empirical design uses DataCo, Favorita, and M5 for distinct operational roles. Table 3 summarizes these roles. DataCo is not treated as an inventory-control dataset; it is used to transfer empirical heterogeneity in execution risk into scenario design. Favorita is the main forecast-to-plan proof-of-concept dataset. M5 is the hierarchy and intermittent-demand robustness dataset.

Table 3: Dataset roles and experimental purpose.

Dataset	Role	Data used	Purpose
DataCo	Execution-risk anchor	Late-delivery outcomes by operational context	Defines data-informed execution-risk scenarios; not treated as an inventory-control dataset.
Favorita	Main proof of concept	Daily store-family retail demand panel	Tests forecast-to-plan conversion and deployable strategy ranking under execution-risk scenarios.
M5	Robustness analysis	Hierarchical and intermittent retail demand	Tests whether the planning-execution gap persists across planning grain and demand sparsity.

DataCo execution-risk anchor. DataCo contains supply-chain fulfillment records with late-delivery outcomes. The analysis extracts late-delivery anchors from context-level percentiles. The current generated anchors range from 0.535 at the low-risk percentile to 0.901 at the severe-risk percentile. Under the additive clipped scenario mapping, these anchors produce execution weights from 0.318 to 0.500. These are sensitivity weights, not exact dollar costs for Favorita or M5.

Favorita proof of concept. Favorita is a daily retail demand-planning panel from Corporacion Favorita. The pipeline treats each store-family pair as a planning unit and uses promotion, holiday, oil-price, transaction, and store-context information where available at planning time. Candidate forecasts include transparent baselines and machine-learning candidates such as LightGBM and XGBoost when dependencies are available (Ke et al. 2017; Chen and Guestrin 2016). The purpose is to test whether forecast-to-plan conversion changes deployable strategy ranking.

M5 robustness design. M5 contains 3,049 items across 10 stores, and the pipeline samples item-store series for quick and medium robustness runs (Makridakis et al. 2022). The IEOM submission uses M5 to evaluate item-store, department-store, and category-store planning grains and to test intermittent-demand buckets. This design checks whether the planning-execution gap persists when demand is sparse or aggregated.

Walmart and Rossmann are not completed experiments in this IEOM draft. They can be used later as optional future robustness panels after the DataCo-Favorita-M5 story is finalized.

5 Results and Discussion

5.1 Numerical Results

Quantifying the planning-execution gap. The strongest baseline validation evidence is the mismatch between the WAPE-best deployable strategy and the operational-loss-best deployable strategy. Across 123 paper-facing evaluation groups, the accuracy-first deployable method differs from the operational-loss-optimal deployable method in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. These two measures validate the central premise that accuracy-first selection and deployable planning selection are not equivalent. Table 4 summarizes the core numerical evidence.

Table 4: Core numerical evidence for the planning-execution gap.

Metric	Current value	Interpretation
Accuracy-first mismatch rate	91.1% of 123 groups	WAPE-best deployable strategy often differs from operational-loss-best deployable strategy.
Mean accuracy-first loss gap	11.169	Average normalized-loss disadvantage of accuracy-first selection when it is not operationally best.
Accuracy-first PEG rate	15.4%	Fraction of periods with positive execution violation for accuracy-first strategies.
Best deployable PEG rate	2.7%	Same metric for the best deployable operational-loss strategy.
Greedy-to-DP gap	0.044	Measures myopic selection cost where greedy and DP are both available.
Mean gap to DP oracle	0.198	Selection suboptimality relative to non-deployable realized-inventory DP information.
Mean gap to perfect oracle	-8.936	Diagnostic total gap; negative value reflects that realized-demand tracking can create high execution burden.
DataCo execution-risk anchors	0.535 to 0.901	Late-delivery anchors used for low to severe execution-risk scenarios.

Ranking shifts under DataCo-informed execution risk. Favorita results provide scenario validation. Global Best remains a strong accuracy-first reference. In the baseline scenario, Global Best has WAPE 0.109 and normalized total loss 1.550, while Operational Ensemble has WAPE 0.119 and normalized total loss 1.422. Under the severe DataCo-informed execution-risk scenario, Global Best remains accurate but has normalized total loss 2.000. Simple Ensemble and Operational Ensemble have lower normalized losses of 1.473 and 1.496, respectively. This does not imply a universal winner. It shows that deployable ranking depends on execution-risk regime.

Robustness under hierarchy and intermittency. M5 provides robustness validation rather than a repetition of Favorita. Table 5 shows that Operational Ensemble is strongest at department-store and category-store grains, while Smoothed Alpha 0.25 is strongest at the item-store grain and across low-, medium-, and high-intermittency buckets. This strengthens the paper because the framework is not designed to make one method always win. Instead, it shows that the preferred deployable mechanism changes with operational context.

Myopic selection and oracle gaps. The generated thesis summary provides algorithmic validation. The mean greedy-to-DP gap is 0.044, indicating that finite-horizon planning improves over one-step greedy selection where both are available. The mean gap to the realized-inventory DP oracle is 0.198, which measures selection suboptimality relative to non-deployable inventory-outcome information. The mean gap to the perfect-demand oracle is negative in the current combined outputs because a realized-demand signal can create high volatility and execution penalties; this confirms that the perfect-demand oracle is a diagnostic reference, not a deployable lower bound.

Table 5: M5 hierarchy and intermittency robustness summary.

Robustness slice	Best deployable method	WAPE	PEG rate	Norm. loss
Hierarchy: category-store	Operational Ensemble	0.159	0.063	1.261
Hierarchy: department-store	Operational Ensemble	0.162	0.100	1.177
Hierarchy: item-store	Smoothed Alpha 0.25	0.368	0.010	1.223
Intermittency: high	Smoothed Alpha 0.25	0.616	0.030	1.284
Intermittency: low	Smoothed Alpha 0.25	0.254	0.000	1.156
Intermittency: medium	Smoothed Alpha 0.25	0.367	0.001	1.168

5.2 Graphical Results

Figure 2 plots the Favorita accuracy versus execution-penalty frontier. The figure shows that WAPE and execution penalty do not move together. Global Best is near the best WAPE region, but other deployable strategies produce much lower execution penalties. The operational implication is that the deployable frontier is not the WAPE-best point.

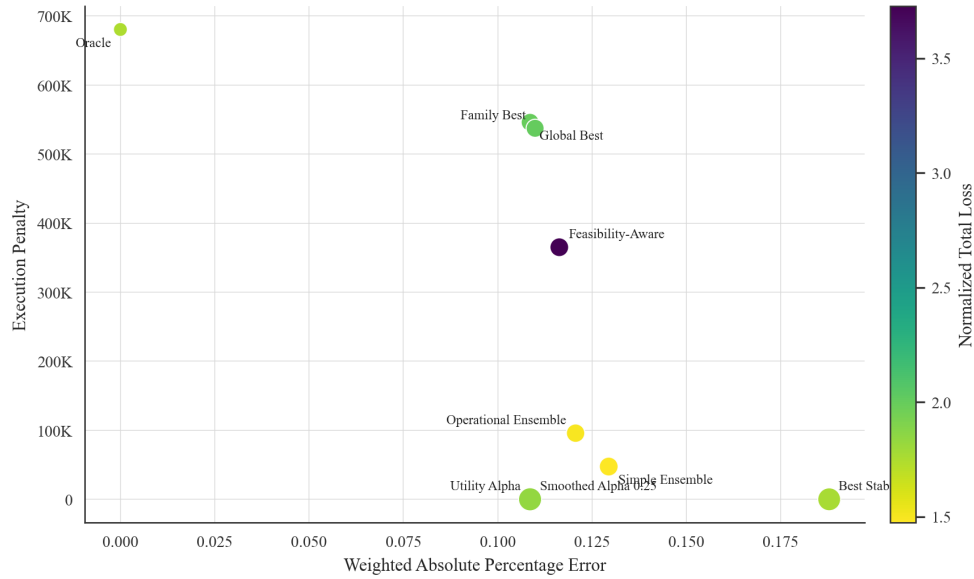


Figure 2: Favorita accuracy and execution penalty frontier.

Figure 3 shows normalized planning loss across DataCo-informed execution scenarios. As execution-risk weight changes, strategy ranking changes. This supports scenario testing as a deployment practice: the model selected for a flexible execution environment may not be the model selected for a constrained execution environment.

5.3 Proposed Improvements

Monitor planning-execution gap rate. The accuracy-first planning-execution gap rate is 15.4%, compared with 2.7% for the best deployable operational-loss strategy. This supports adding PEG rate to routine model-deployment monitoring.

Avoid WAPE-only model selection. The accuracy-first deployable method differs from the operational-loss-optimal deployable method in 91.1% of paper-facing evaluation groups. Planners should treat WAPE as diagnostic, not as the only deployment criterion.

Use execution-risk scenario testing. Favorita rankings change between baseline and severe DataCo-informed scenarios. Before rollout, planning teams should test candidate strategies under plausible execution-capacity regimes.

Use smoothing and ensembles selectively. Smoothing can reduce execution burden, but it can also increase inventory exposure when adaptation is too slow. Ensembles can reduce abrupt model-specific movements, but they may sacrifice

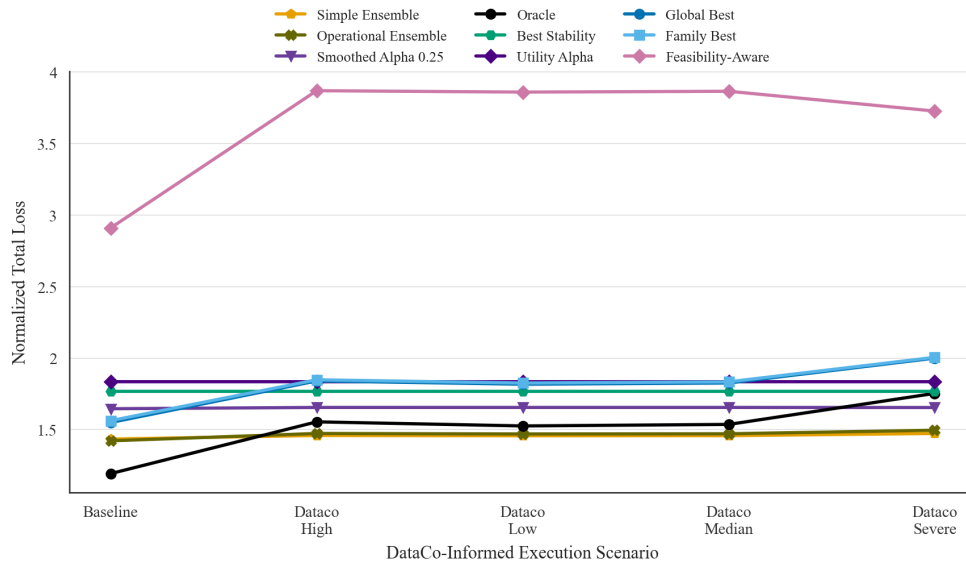


Figure 3: Favorita normalized planning loss by DataCo-informed execution scenario.

WAPE. These methods should be selected by stated operational tradeoffs rather than by a universal rule.

Use DP or budgeted DP when switching governance matters. The greedy-to-DP gap shows that one-step selection can be myopic. Budgeted DP is useful when planners impose hard limits on model switching, approvals, or monitoring changes.

5.4 Validation

5.4.1 Baseline Validation

Baseline validation asks whether the accuracy-first strategy is the same strategy a planner would deploy after feasibility adjustment. Accuracy-first methods remain necessary baselines because they represent the conventional forecasting benchmark. The 91.1% mismatch rate and the PEG-rate comparison show that WAPE-best selection often differs from operational-loss-best deployable selection. The paper therefore does not claim that feasibility-aware strategies always dominate; it validates that the two selection criteria answer different operational questions.

5.4.2 Scenario Validation

Scenario validation asks whether execution pressure changes strategy rankings. DataCo is used to construct execution-risk sensitivity scenarios from empirical late-delivery behavior. The Favorita baseline-to-severe shift shows that a strategy that is attractive when execution pressure is low can become less attractive when execution-capacity violations are weighted more heavily. This validates the planning-execution gap as a scenario-dependent deployment issue rather than a fixed property of one model.

5.4.3 Algorithmic Validation

Algorithmic validation asks whether the decision layer matters after the feasibility terms are defined. The greedy selector is a myopic one-period selector, while DP is a finite-horizon selector that accounts for future transition effects. The greedy-to-DP gap measures the cost of myopic selection. Budgeted DP adds a governance constraint by limiting the number of model switches; it is relevant when model changes require monitoring, approval, or infrastructure updates.

The benchmark gaps serve different validation purposes. The DP-oracle gap measures selection suboptimality relative to a non-deployable realized-inventory DP benchmark over the same forecast candidates. The perfect-demand reference gap is a diagnostic total gap that combines forecast uncertainty and feasibility effects. A perfect-demand signal does not automatically imply lower operational loss because exact demand tracking can create high planning volatility and execution violations.

5.4.4 Robustness Validation

Robustness validation asks whether the framework remains informative outside the main Favorita proof of concept. Favorita provides a context-rich retail demand panel with store-family planning units. M5 tests hierarchy and intermittency by changing planning grain and demand sparsity. The observed changes in preferred deployable method across M5 slices support the thesis that feasibility-constrained planning is a tradeoff surface, not a single-method leaderboard.

6 Conclusion

This paper reframes operational demand planning as feasibility-constrained model selection. The planning-execution gap was quantified by comparing accuracy-first model selection with operational-loss-optimal deployable planning selection. Across 123 paper-facing evaluation groups, the accuracy-first deployable method differs from the operational-loss-best deployable method in 91.1% of cases. The accuracy-first planning-execution gap rate is 15.4%, while the best deployable operational-loss strategy has a gap rate of 2.7%.

Objective 1 was addressed by defining a forecast-to-plan evaluation layer that reports WAPE separately from inventory exposure, volatility, switching burden, and execution violations. Objective 2 was addressed by defining planning-execution gap rate. Objective 3 was addressed by using DataCo execution records to construct data-informed execution-risk scenarios. Objective 4 was addressed by comparing accuracy-first, ensemble, smoothing, greedy, DP, and budgeted-DP deployable strategies. Objective 5 was addressed through Favorita and M5 validation and through evidence-linked planning recommendations.

The results show that execution-risk scenarios can change deployable strategy ranking and that hierarchy and intermittency can change the preferred mechanism. The practical implication is direct: planners should evaluate deployment feasibility before model rollout. Forecast accuracy remains important, but it should be interpreted as a diagnostic metric rather than as the sole deployment objective.

Acknowledgements

Acknowledgement placeholder. Replace this paragraph with funding, advisor, or institutional acknowledgements before submission if applicable.

Biography

Author biography placeholder. Replace this paragraph with a biography for each author before submission. IEOM guidance limits each biography to 250 words.

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